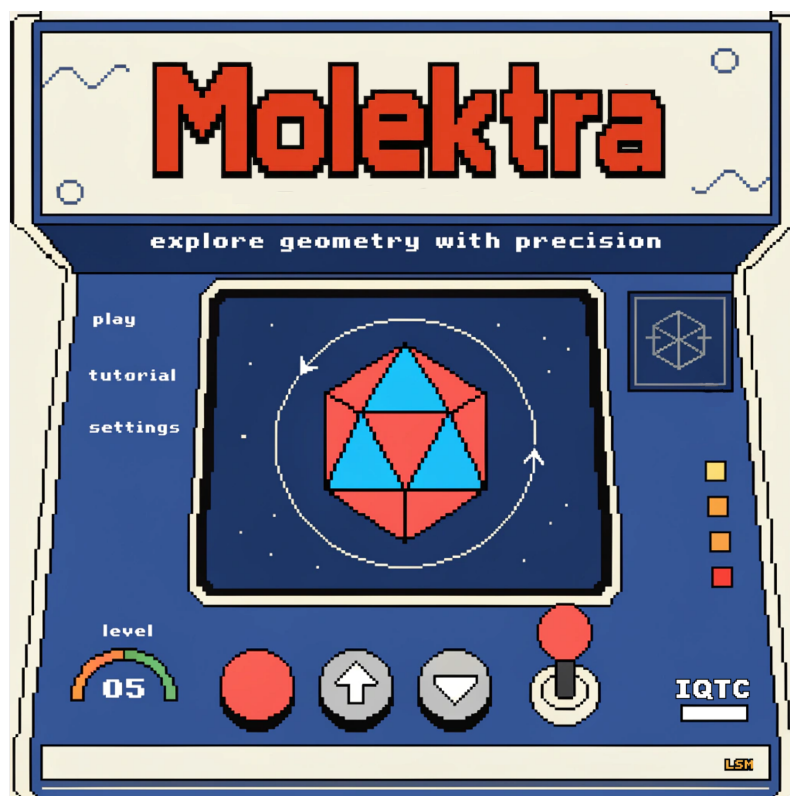


Molektra



Molecular visualization tool

User manual version 1.0

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1. Introduction

The Continuous Shape Measures (CShM) and Continuous Symmetry Measures (CSM) are powerful methodologies that have been widely used in computational chemistry and coordination chemistry over the last 20 years to quantify deviations from ideal molecular geometries and symmetries. These approaches provide a rigorous numerical framework for comparing real molecular structures against ideal polyhedral or symmetry reference models, allowing researchers to analyze structural distortions in a precise and systematic way. Despite their usefulness, the practical application of these methods has traditionally relied on command-line programs, which often require manual preparation of input files, specialized syntax and computer skills, and interpretation of non-visual numerical outputs. Such limitations can make the analysis process difficult for user, less intuitive, and less accessible to the broader audience.

To address these challenges, the present software has been developed as an interactive environment that integrates molecular visualization with symmetry analysis and continuous shape measures. By integrating CShM and CSM methodologies within a visual and user-friendly platform, this software allows the visualization, atom selection, geometric measurements, symmetry determination, and the interpretation of the result for a given molecular structure. The software combines real-time 3D molecular rendering with automated analysis tools, allowing the user to perform advanced structural studies more efficiently while reducing the complexity associated with traditional console-based approaches.

2. Software

2.1. Software Setup

To ensure that the graphical interface and all its scientific dependencies work correctly, the software must be downloaded in the version corresponding to your operating system (Windows or macOS). If you are using macOS, please note that a different version is required depending on whether your device is equipped with an M1/M2 (Apple Silicon) or Intel chip. Additionally, macOS security defaults do not usually allow you to open downloaded software. To bypass macOS security blocks on

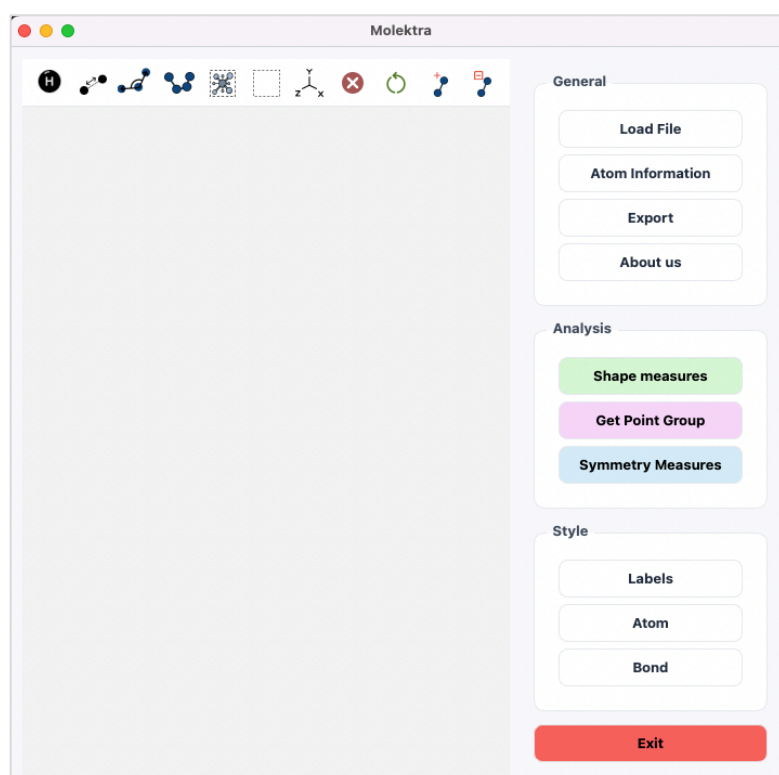
downloaded files, right-click (or Control-click) the file, select Open, and click Open Anyway.

2.2. Calling Molektra

Double-click the executable file.

3. Molektra Main Interface

If installation has been successful, you should encounter the Molektra main window, which contains the following options



3.1 Main Window Overview

When the program starts, the main graphical interface is displayed. This window provides access to all visualization, measurement, and symmetry-analysis tools available in the software.

The interface is divided into two main sections:

- 3D Visualization Panel with a toolbar (left side): Displays the loaded molecular structure and allows interactive manipulation of the model.

- Control Panel (right side): Contains the core tools and settings, grouped into distinct sections, alongside an application exit control:

-General: Encompasses administrative functions of the program. This includes loading files, displaying information for selected atoms, exporting results or 3D visualizations, and accessing the "About us" information.

-Analysis: Contains the core computational tools of the software. From this section, users can perform Continuous Shape Measures (CShM), determine the molecular point group and execute symmetry analysis.

-Style: Allows users to customize the visual rendering of the 3D model, providing access settings to modify labels, atoms representations and bond styles.

-Exit: Control button to safely close the application.

3.2. 3D Visualization Area:

The large panel on the left serves as the interactive molecular viewer. After loading a structure file, the molecule will be rendered in this region. The viewer supports full interactive manipulation, including:

- Rotation, translation and zooming of the molecular structure
- Single and multiple atom selection
- Visualization of geometric measurements
- Display of coordination polyhedra or symmetry elements.

The toolbar

At the top of the visualization panel, a dedicated toolbar provides quick access to essential features. You can move your mouse over any icon to display a tooltip with its description. The available tools can be classified as:

Visualization Options

-Show/Hide hydrogens ( [h]): Toggles the visibility of hydrogen atoms in the molecular visualization. Designed to improve clarity for larger molecules.

-Show/Hide Axes ( [x]): Displays or hides the Cartesian coordinates axes in the viewer. These axes help orient the molecular structure in 3D space.

Measurements

- Measure distance () [*d*]: Measures the distance between two selected atoms.
- Measure angle () [*a*]: Calculates the bond angle between three selected atoms.
- Measure dihedral angle (): Calculates the torsion angle between four selected atoms.

Selection tools

- Isolate coordination environment () [*c*]: Temporarily hides the rest of the structure to focus exclusively on the immediate coordination sphere of the selected central atom.
- Area selection (): Enables rectangular or free-hand (lasso) selection of multiple atoms simultaneously.

Structure Editing and Reset

- Add bond () [*b*]: Creates a bond between two selected atoms.
- Delete bond () [*Shift* + *b*]: Removes a bond between two selected atoms.
- Delete atom () [*Delete*]: Removes the currently selected atom(s) from the visualization
- Reset () [*Ctrl/Cmd* + *z*]: Restores the molecular structure to its original state.

3.3. Control Panel

The right-side panel contains all action buttons and customization tools, organized into three main categories: General, Analysis, and Style.

General

Load File: Loads a molecular structure file into the viewer. Supported coordinate formats include currently cartesian coordinates files, (file.xyz) crystallographic information files (file.CIF) and Protein Data Bank files (file.pdb). After loading, atoms and bonds are automatically rendered.

Atom Information: Displays detailed information about selected atoms, including the atom label, element type, electronegativity, coordination number and other relevant chemical properties.

Export: Allows users to export calculated data and analysis results to external files in a CSV format. Additionally, users can export high-quality images of the current molecular visualization in TIFF, PNG or JPEG formats, offering customizable resolution and background options.

About us: Displays information about the software, including developers and relevant references.

Analysis

Calculate Shape: Computes the molecular shape descriptors using the SHAPE methodology. This option compares the loaded geometry against ideal reference polyhedra and provides with the Continuous Shape Measures (CShM) for all polyhedra with the same number of vertices as defined by the user.

Get Point Group: Determines the molecular point group symmetry of the structure using symmetry-analysis libraries. The detected point group is displayed in a results window or console output.

Symmetry Measures: Calculates continuous symmetry measures (CSM) for the loaded structure. These values quantify how closely the molecule approaches ideal symmetry.

Style

Labels: Allows users to toggle and customize atom labels within the 3D viewer.

Atom: Provides setting to adjust the visual representation of atoms, such as modifying their color schemes or radii.

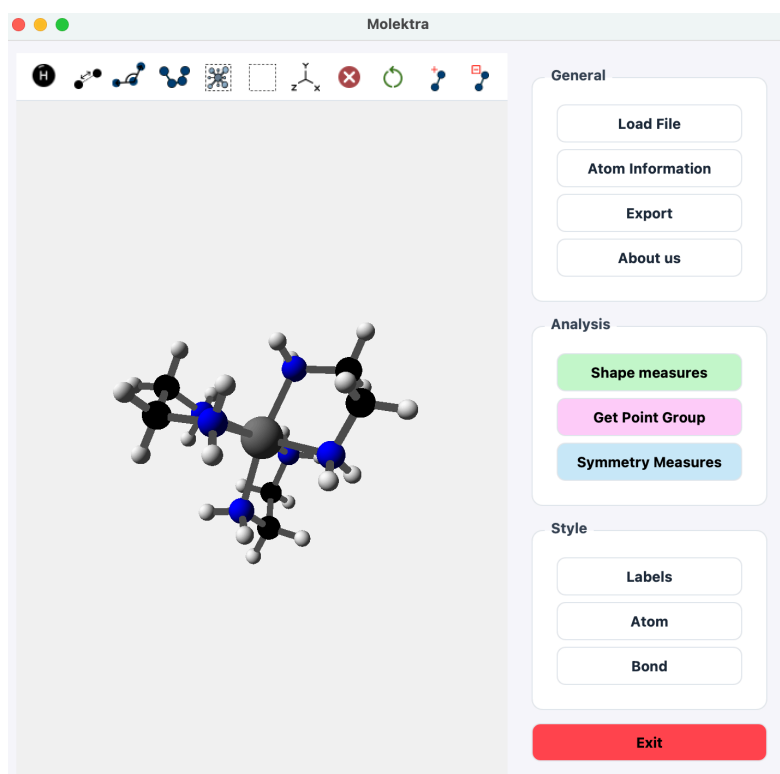
Bond: Provides settings to modify the visual style of the bond, including adjustments to cylinder thickness and color.

Application Controls

Exit: Safely closes the application.

4. Determination of the CShM values

In this section, we will work out the determination of the CShM values for the $[\text{Cu}(\text{en})_3]$ system. The typical workflow is as follows:



1. Launch the software.

Open *Molektra* to access the main interface.

2. Load the molecule.

Click the **Load File** button to open the file selection dialog. The software installation includes a folder with example files; for this case navigate to the *files_xyz* folder, select the *Cuen3.xyz* file, and click Open.

(Note: Alternatively, you can directly drag and drop the file into the 3D visualization screen).

3. Inspect the structure.

The 3D viewer will now display the loaded molecule, as shown in the figure below.

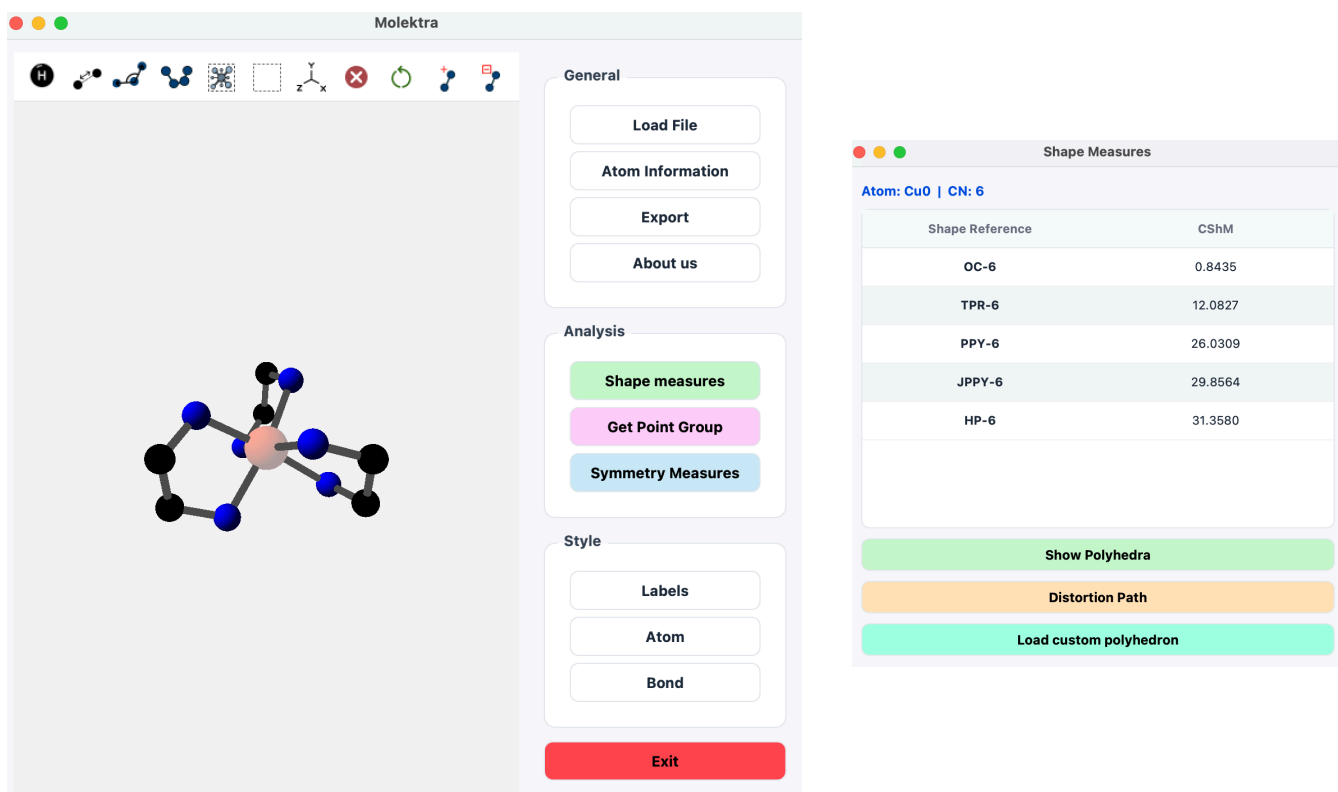
4. Run the shape analysis.

Select the central Copper (Cu) atom by clicking on it in the viewer; the selected atom will be highlighted in red. Next, click the **Shape measures** button in the

Analysis panel. The program automatically detects that the Cu is bound to six nitrogen atoms, will compute the CShM for all polyhedra with 6 vertices.

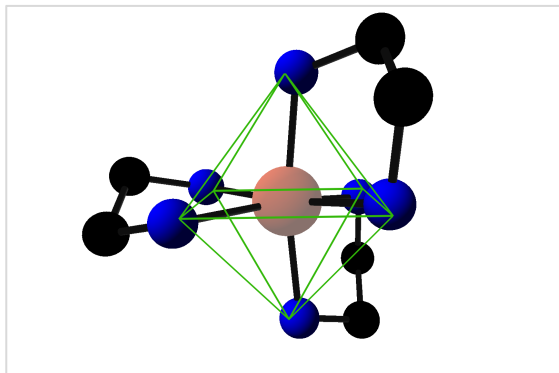
5. Interpret the results.

A pop-up “Shape Measures” window will appear, displaying a table with the CShM values for the five polyhedron with six vertices included in the SHAPE code. From the displayed values, user can clearly see that CShM(OC-6) has the lowest value, indicating that the Cu coordination environment is closest to an ideal octahedron.



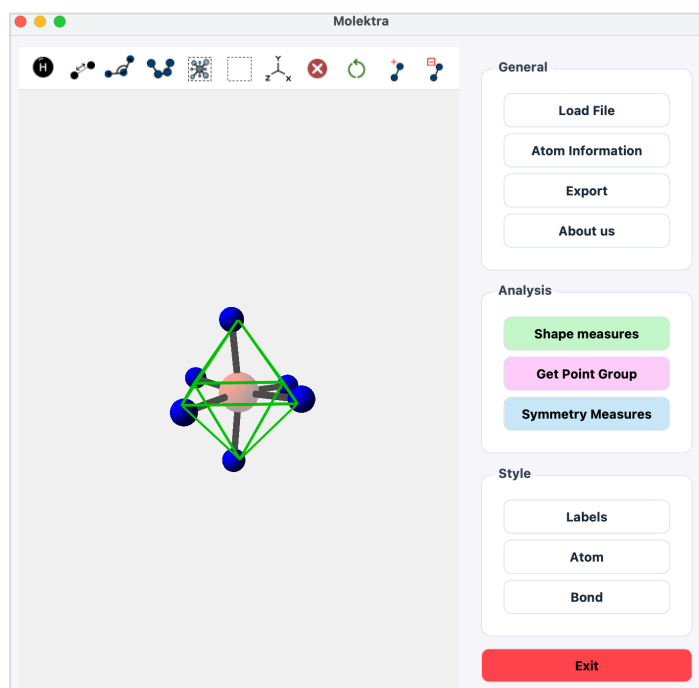
6. Visualize the ideal polyhedron

You can overlap the different ideal reference polyhedra with the molecule by using the **Show Polyhedra** button. From the different options, the user can click and select a given polyhedron and overlap it in the best possible orientation with the selected atoms. In the figure below this is illustrated for the octahedron. The user can customize the color of the ideal polyhedron using the **Set polyhedron color** button. Additionally, you can load a custom reference polyhedron from an external .xyz file to compare it directly against your structure.



7. Adjust visualization options.

To improve clarity, *Molektra* provides several visualization tools in the top toolbar. For instance, you can use **Show/Hide Hydrogen** [h] to exclude hydrogen atoms from the display (as seen in previous figures). Alternatively, you can use **Isolate Coordination Environment** [c] to temporarily hide all atoms excepts those in the first coordination sphere. This isolated view is illustrated in the figure below.



8. Export data and images.

At any point during the analysis, the user can save the results using the **Export** button. This saves the numerical measurements in a .csv file, including the labels of all evaluated polyhedra and their corresponding CShM values. If desired, the user can also export a high-quality image of the molecule overlaid with the selected polyhedron.

5. Point Group and Continuous Symmetry Measures

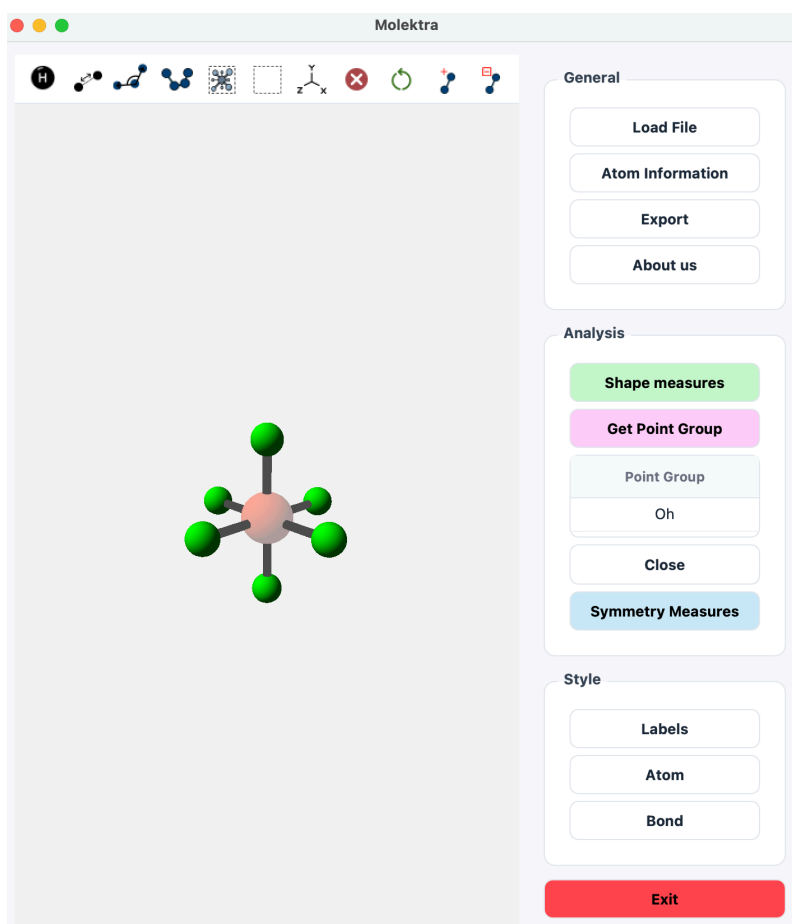
Molektra provides a comprehensive set of tools for symmetry analysis. In this example, we will determine the point group and perform a Continuous Symmetry Measures (CSM) analysis for a $[\text{MnCl}_6]^{3-}$.

1. Load the molecule.

As in the previous workflow, start by selecting and loading your molecular structure file into the 3D viewer.

2. Determine the Point Group.

From the Analysis panel in the main window, click the **Get Point Group** button. This button displays the automatically detected molecular point group. The inset can be closed using the **Close** button. The determination of the O_h point group for the $[\text{MnCl}_6]^{3-}$ anion is illustrated below.

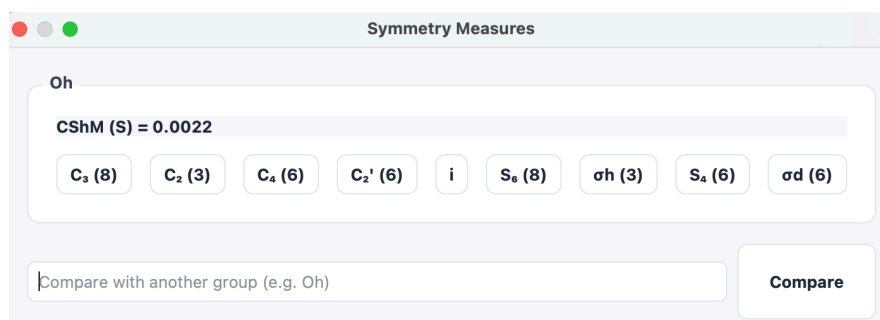


3. Run the Symmetry Measures analysis.

To perform a detailed CSM analysis, click the Symmetry Measures button. This will open a dedicated results windows containing:

- The overall CSM value for the detected Point Group.
- A detailed list of all individual symmetry operations present within that group.
- A comparison tool at the bottom to evaluate the structure against any other point group of interest.

For the $[\text{MnCl}_6]^{3-}$ case discussed above, this menu appears as illustrated below:



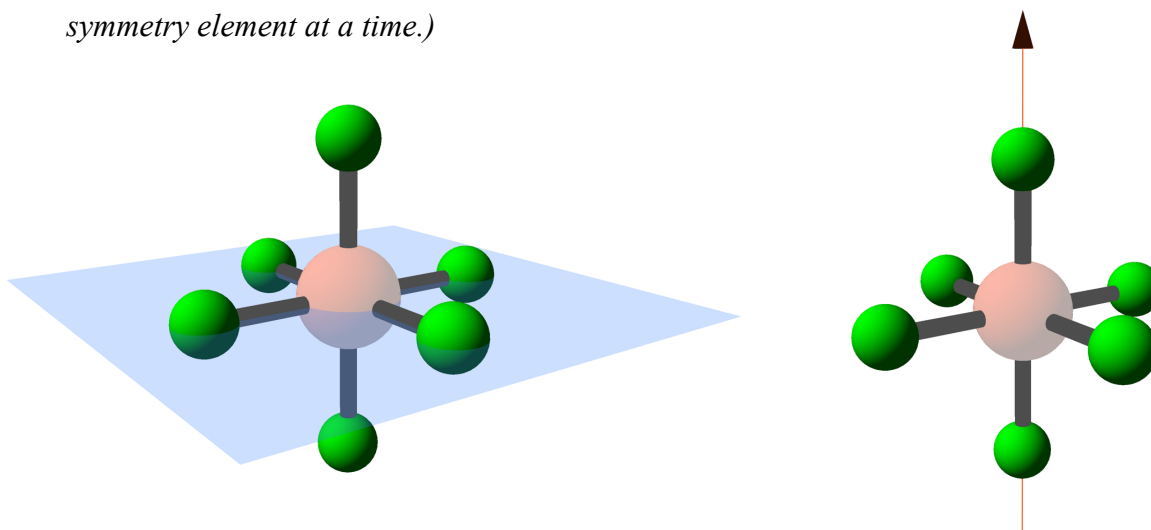
4. Inspect symmetry elements.

The results menu provides the CSM for the point group alongside all its symmetry operations. For each operation, the total number of symmetry elements is displayed in parentheses.

5. Visualize symmetry elements.

You can select any symmetry operation from the list to superimpose its corresponding symmetry element (such as rotation axes, mirror planes, or inversion center) directly onto the 3D molecule. Below are illustrated examples of how the C_4 axis or the σ_h reflection plane look after being selected.

(Note: The current version of Molektra supports the visualization of only one symmetry element at a time.)



6. Keyboard Shortcuts Summary

For a faster workflow, *Molektra* supports several keyboard shortcuts for its most common tools.

Category	Action	Shortcut
Visualization	Show/hide Hydrogens	[h]
	Show/Hide Axes	[x]
	Isolate Coordination Environment	[c]
Measurements	Measure Distance	[d]
	Measure Angle	[a]
Selection & Editing	Expand Selection	[Ctrl/Cmd + E]
	Add Bond	[b]
	Delete Bond	[Shift + b]
	Delete Atom(s)/Selection	[Delete]
General	Reset Structure	[Ctrl/Cmd + z]

7. List of built-in reference polyhedra

Vertices	Code	Label	Shape	Symmetry
2	1	L-2	Linear	$D_{\infty h}$
	2	vT-2	Divacant tetrahedron	C_{2v}
	3	vOC-2	Tetravacant octahedron	C_{2v}
3	1	TP-3	Trigonal planar	D_{3h}
	2	vT-3	Pyramid	C_{3v}
	3	fac-vOC-3	fac-Trivacant octahedron	C_{3v}
	4	mer-vOC-3	mer-Trivacant octahedron (T-shape)	C_{2v}

4	1	SP-4	Square	D _{4h}
	2	T-4	Tetrahedron	T _d
	3	SS-4	Seesaw or sawhorse	C _{2v}
	4	vTBPY-4	Axially vacant trigonal bipyramid	C _{3v}
5	1	PP-5	Pentagon	D _{5h}
	2	vOC-5	Vacant octahedron	C _{4v}
	3	TBPY-5	Trigonal bipyramid	D _{3h}
	4	SPY-5	Square pyramid	C _{4v}
	5	JTBPY-5	Johnson trigonal bipyramid	D _{3h}
6	1	HP-6	Hexagon	D _{6h}
	2	PPY-6	Pentagonal pyramid	C _{5v}
	3	OC-6	Octahedron	O _h
	4	TPR-6	Trigonal prism	D _{3h}
	5	JPPY-5	Johnson pentagonal pyramid	C _{5v}
7	1	HP-7	Heptagon	D _{7h}
	2	HPY-7	Hexagonal pyramid	C _{6v}
	3	PBPY-7	Pentagonal bipyramid	D _{5h}
	4	COC-7	Capped octahedron	C _{3v}
	5	CTPR-7	Capped trigonal prism	C _{2v}
	6	JPBPY-7	Johnson pentagonal bipyramid	D _{5h}
	7	JETPY-7	Elongated triangular pyramid	C _{3v}
8	1	OP-8	Octagon	D _{8h}
	2	HPY-8	Heptagonal pyramid	C _{7v}
	3	HBPY-8	Hexagonal bipyramid	D _{6h}
	4	CU-8	Cube	O _h
	5	SAPR-8	Square Antiprism	D _{4d}
	6	TDD-8	Triangular dodecahedron	D _{2d}
	7	JGBF-8	Johnson - Gyrobifastigium	D _{2d}
	8	JETBPY-8	Johnson - Elongated triangular bipyramid	D _{3h}
	9	JBTP-8	Johnson - Biaugmented trigonal prism	C _{2v}
	10	BTPR-8	Biagumented trigonal prism	C _{2v}

	11	JSD-8	Snub disphenoid	D _{2d}
	12	TT-8	Triakis tetrahedron	T _d
	13	ETBPY-8	Eloganted trigonal bipyramid	D _{3h}
9	1	EP-9	Enneagon	D _{9h}
	2	OPY-9	Octagonal pyramid	C _{8v}
	3	HPY-9	Heptagonal bipyramid	D _{7h}
	4	JTC-9	Triangular cupola	C _{3v}
	5	JCCU-9	Capped Cube (Elongated square pyramid)	C _{4v}
	6	CCU-9	Capped Cube	C _{4v}
	7	JCSAPR-9	Capped sq. Antiprism (Gyroelongated square pyramid)	C _{4v}
	8	CSAPR-9	Capped square Antiprism	C _{4v}
	9	JTCTPR-9	Tricapped trigonal prism (J51)	D _{3h}
	10	TCTPR-9	Tricapped trigonal prism	D _{3h}
	11	JTDIC-9	Tridiminshed icosahedron	C _{3v}
	12	HH-9	Hula-hoop	C _{2v}
	13	MFF-9	Muffin	C _s
10	1	DP-10	Decagon	D _{10h}
	2	EPY-10	Enneagonal pyramid	C _{9v}
	3	OBPY-10	Octagonal bipyramid	D _{8h}
	4	PPR-10	Pentagonal prism	D _{5h}
	5	PAPR-10	Pentagonal antiprism	D _{5d}
	6	JBCCU-10	Bicapped cube	D _{4h}
	7	JBCSAPR-10	Bicapped square antiprism	D _{4d}
	8	JMBIC-10	Metabidiminshed icosahedron	C _{2v}
	9	JATDI-10	Augmented tridiminshed icosahedron	C _{3v}
	10	JSPC-10	Sphenocorona	C _{2v}
	11	SDD-10	Staggered dodecahedron	D ₂
	12	TD-10	Tetradecahedron	C _{2v}
	13	HD-10	Hexadecahedron	D _{4h}
11	1	HP-11	Hendecagon	D _{11h}

	2	DPY-11	Decagonal pyramid	C_{10v}
	3	EBPY-11	Enneagonal bipyramid	D_{9h}
	4	JCPPR-11	Capped pent. Prism	C_{5v}
	5	JCAPR-11	Capped pent. Antiprism	C_{5v}
	6	JAPPR-11	Augmented pentagonal prism	C_{2v}
	7	JASPC-11	Augmented sphenocorona	C_s
12	1	DP-12	Dodecagon	D_{12h}
	2	HPY12	Hendecagonal pyramid	C_{11v}
	3	DBPY-12	Decagonal bipyramid	D_{10h}
	4	HPR-12	Hexagonal prism	D_{6h}
	5	HAPR-12	Hexagonal Antiprism	D_{6d}
	6	TT-12	Truncated tetrahedron	T_d
	7	COC-12	Cuboctahedron	O_h
	8	ACOC-12	Anticuboctahedron	D_{3h}
	9	IC-12	Icosahedron	I_h
	10	JSC-12	Square cupola	C_{4v}
	11	JEPBPY-12	Elongated pentagonal bipyramid	D_{6h}
	12	JBAPPR-12	Biaugmented pentagonal prism	C_{2v}
	13	JSPMC-12	Sphenomegacorona	C_s

8. How to cite

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